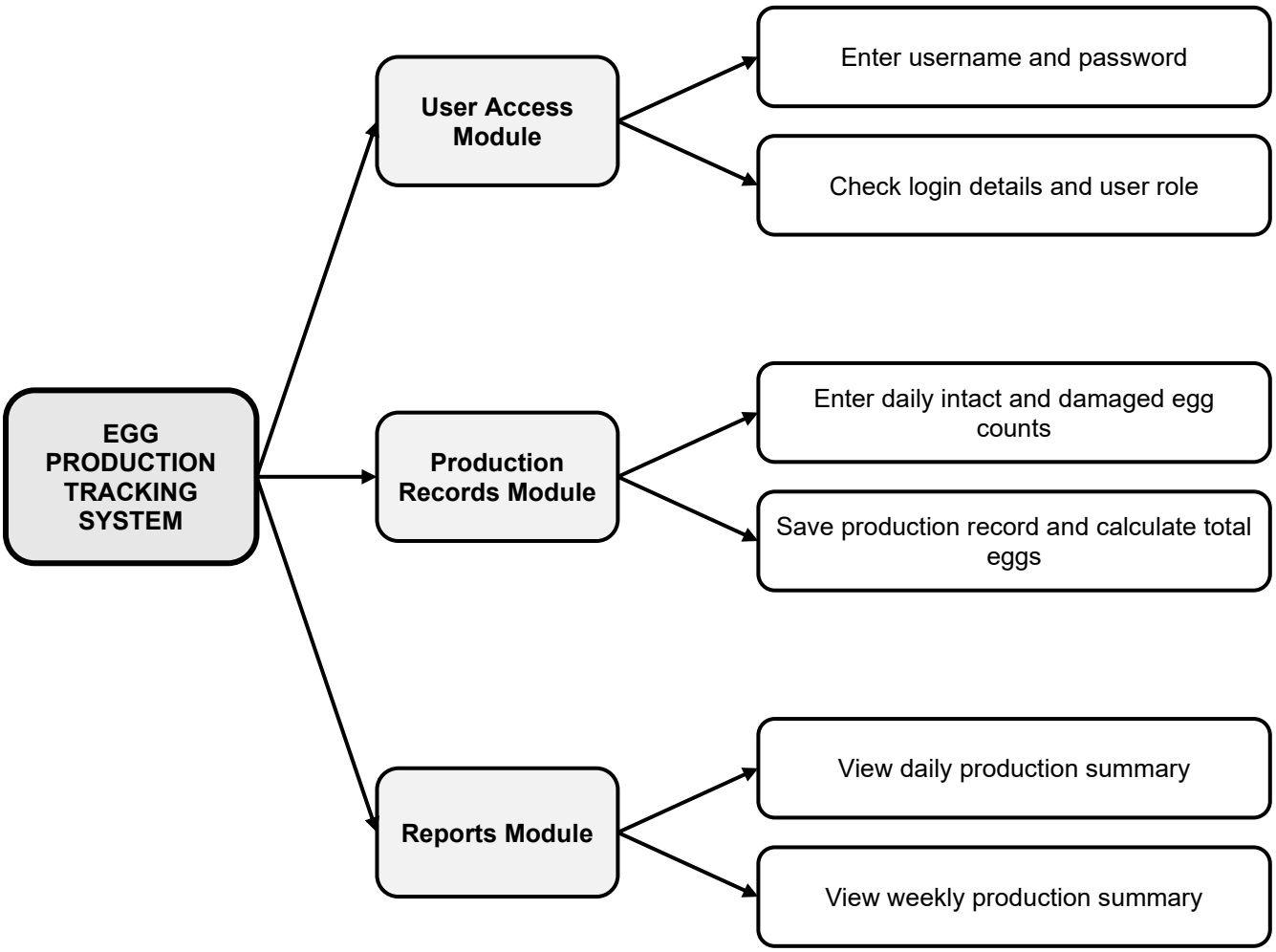
Model Diagram

The diagram below shows how the system is divided into smaller modules. Each module is responsible for a main task in recording and viewing egg production.



Explanation of the Modules

1. User Access Module

This module allows the farm owner and the record-keeping worker to access the system using their login details. It checks the username and password and identifies the user's role. This helps prevent people who are not allowed to use the system from accessing the records.

2. Production Records Module

This is the main module of the system. It allows the worker to enter the number of intact eggs and damaged eggs collected each day. It saves the information as a production record and calculates the total eggs collected. The damaged eggs remain recorded separately, as the farm currently handles them separately.

3. Reports Module

This module uses the saved production records to show daily and weekly egg production summaries. It helps the owner check production figures over time without having to calculate totals manually from the counter book.

Cohesion

Cohesion means how well the tasks inside one module belong together. The system aims for functional cohesion, which is the best type shown in the lecturer’s notes. This means each module has one clear job.

In the diagram, each module in the middle column is connected only to the tasks it performs on the right, and every task sits under one module only. The User Access Module is connected to “Enter username and password” and “Check login details and user role”, and both tasks serve one purpose, which is letting the right person log in. The Production Records Module is connected to “Enter daily intact and damaged egg counts” and “Save production record and calculate total eggs”, and both tasks are about capturing the day’s egg production. The Reports Module is connected to “View daily production summary” and “View weekly production summary”, and both tasks are about displaying production summaries. No task in the diagram looks out of place in its module, so the tasks in each module support its main job.

Coupling

Coupling means how much one module depends on another module. The system aims for data coupling, which is the best type shown in the lecturer’s notes. This means modules share only the information they need.

In the diagram, the User Access, Production Records and Reports modules each branch separately from the Egg Production Tracking System box, and there are no arrows running directly between the modules. This shows that none of them controls or changes the inner workings of another. The only link between them is data: the Reports Module uses the saved production records to prepare the daily and weekly summaries shown on the right of the diagram. It does not need to control how the Production Records Module works or change its internal data directly. Keeping the modules separate in this way makes the system easier to understand and update.

Overall, the system is divided into modules with clear responsibilities. This gives each module good functional cohesion and keeps the connection between modules simple through data coupling.